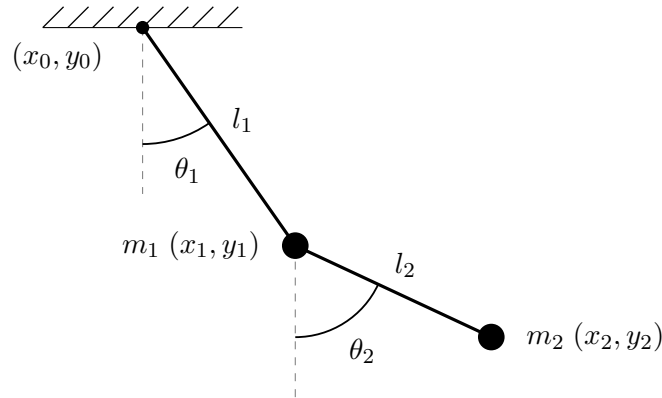


Euler–Lagrange derivation (2-arm pendulum)

$$L = T - V$$

Setup

Pivot at the origin. Generalized coordinates: the absolute angles θ_1, θ_2 . Screen convention: $(0, 0)$ is the top-left corner, so y increases downward.



Positions

$$x_1 = x_0 + l_1 \sin \theta_1 \quad y_1 = y_0 + l_1 \cos \theta_1$$

$$x_2 = x_1 + l_2 \sin \theta_2 \quad y_2 = y_1 + l_2 \cos \theta_2$$

Velocities

$$\dot{x}_1 = l_1 \cos \theta_1 \dot{\theta}_1 \quad \dot{y}_1 = -l_1 \sin \theta_1 \dot{\theta}_1$$

$$v_1 = \sqrt{\dot{x}_1^2 + \dot{y}_1^2} \quad v_1^2 = \dot{x}_1^2 + \dot{y}_1^2$$

$$\dot{x}_2 = \dot{x}_1 + l_2 \cos \theta_2 \dot{\theta}_2 \quad \dot{y}_2 = \dot{y}_1 - l_2 \sin \theta_2 \dot{\theta}_2$$

$$v_2 = \sqrt{\dot{x}_2^2 + \dot{y}_2^2} \quad v_2^2 = \dot{x}_2^2 + \dot{y}_2^2$$

Kinetic energy

$$T = \frac{1}{2}mv^2 \quad \Longrightarrow \quad T = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

Potential energy

$$V = mgh, \quad h_1 = -y_1, \quad h_2 = -y_2$$

$$V = g(m_1h_1 + m_2h_2)$$

$$V = -g(m_1y_1 + m_2y_2)$$

Lagrangian

$$L = T - V$$

$$L = \frac{1}{2}(m_1 v_1^2 + m_2 v_2^2) - g(m_1 h_1 + m_2 h_2)$$

$$L = \frac{1}{2} \left(m_1 (\dot{x}_1^2 + \dot{y}_1^2) + m_2 (\dot{x}_2^2 + \dot{y}_2^2) \right) - g(m_1 y_1 + m_2 y_2)$$

Substituting the velocities:

$$\begin{aligned} L &= \frac{1}{2} m_1 \left((\cos \theta_1 l_1 \dot{\theta}_1)^2 + (-\sin \theta_1 l_1 \dot{\theta}_1)^2 \right) \\ &\quad + \frac{1}{2} m_2 \left((\dot{x}_1 + \cos \theta_2 l_2 \dot{\theta}_2)^2 + (\dot{y}_1 - \sin \theta_2 l_2 \dot{\theta}_2)^2 \right) \\ &\quad + g(m_1 y_1 + m_2 y_2) \end{aligned}$$

$$\begin{aligned} L &= \frac{1}{2} m_1 \left((\cos \theta_1 l_1 \dot{\theta}_1)^2 + (-\sin \theta_1 l_1 \dot{\theta}_1)^2 \right) \\ &\quad + \frac{1}{2} m_2 \left((\cos \theta_1 l_1 \dot{\theta}_1 + \cos \theta_2 l_2 \dot{\theta}_2)^2 + (-\sin \theta_1 l_1 \dot{\theta}_1 - \sin \theta_2 l_2 \dot{\theta}_2)^2 \right) \\ &\quad + g(m_1(y_0 + \cos \theta_1 l_1) + m_2(y_1 + \cos \theta_2 l_2)) \end{aligned}$$

Expanding the squares:

$$\begin{aligned} L &= \frac{1}{2} m_1 \left(\cos^2 \theta_1 l_1^2 \dot{\theta}_1^2 + \sin^2 \theta_1 l_1^2 \dot{\theta}_1^2 \right) \\ &\quad + \frac{1}{2} m_2 \left(\cos^2 \theta_1 l_1^2 \dot{\theta}_1^2 + 2 \cos \theta_1 l_1 \dot{\theta}_1 \cos \theta_2 l_2 \dot{\theta}_2 + \cos^2 \theta_2 l_2^2 \dot{\theta}_2^2 \right. \\ &\quad \left. + \sin^2 \theta_1 l_1^2 \dot{\theta}_1^2 + 2 \sin \theta_1 l_1 \dot{\theta}_1 \sin \theta_2 l_2 \dot{\theta}_2 + \sin^2 \theta_2 l_2^2 \dot{\theta}_2^2 \right) \\ &\quad + g(m_1(y_0 + \cos \theta_1 l_1) + m_2(y_1 + \cos \theta_2 l_2)) \end{aligned}$$

Grouping:

$$\begin{aligned} L &= \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1) \\ &\quad + \frac{1}{2} m_2 (l_1^2 \dot{\theta}_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1) + l_2^2 \dot{\theta}_2^2 (\cos^2 \theta_2 + \sin^2 \theta_2) \\ &\quad + 2 l_1 \dot{\theta}_1 l_2 \dot{\theta}_2 (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2)) \\ &\quad + g(m_1(y_0 + \cos \theta_1 l_1) + m_2(y_1 + \cos \theta_2 l_2)) \end{aligned}$$

Using $\cos^2 + \sin^2 = 1$ and $\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 = \cos(\theta_1 - \theta_2)$:

$$\begin{aligned} L &= \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 \\ &\quad + \frac{1}{2} m_2 \left(l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right) \\ &\quad + g(m_1(y_0 + l_1 \cos \theta_1) + m_2(y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2)) \end{aligned}$$

Euler–Lagrange

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0$$

For θ_1

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1} = 0$$

First $\partial L / \partial \theta_1$:

$$\begin{aligned} \frac{\partial L}{\partial \theta_1} &= \frac{\partial}{\partial \theta_1} \left[\frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 \right. \\ &\quad + \frac{1}{2} m_2 \left(l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right) \\ &\quad \left. + g (m_1 (y_0 + l_1 \cos \theta_1) + m_2 (y_0 + l_1 \cos \theta_1 + l_2 \cos \theta_2)) \right] \\ \frac{\partial L}{\partial \theta_1} &= \frac{1}{2} m_2 \left(2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (-\sin(\theta_1 - \theta_2)) \right) + g (m_1 l_1 (-\sin \theta_1) + m_2 l_1 (-\sin \theta_1)) \\ \frac{\partial L}{\partial \theta_1} &= -m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) - g l_1 (m_1 + m_2) \sin \theta_1 \end{aligned}$$

Then $\partial L / \partial \dot{\theta}_1$:

$$\begin{aligned} \frac{\partial L}{\partial \dot{\theta}_1} &= m_1 l_1^2 \dot{\theta}_1 + \frac{1}{2} m_2 \left(2 l_1^2 \dot{\theta}_1 + 2 l_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right) \\ \frac{\partial L}{\partial \dot{\theta}_1} &= m_1 l_1^2 \dot{\theta}_1 + m_2 l_1 \left(l_1 \dot{\theta}_1 + l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right) \\ \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) &= m_1 l_1^2 \ddot{\theta}_1 \\ &\quad + m_2 l_1 \left(l_1 \ddot{\theta}_1 + l_2 \left(\ddot{\theta}_2 \cos(\theta_1 - \theta_2) + \dot{\theta}_2 (-\sin(\theta_1 - \theta_2)) (\dot{\theta}_1 - \dot{\theta}_2) \right) \right) \end{aligned}$$

Combining and simplifying:

$$\begin{aligned} m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1 \left(l_1 \ddot{\theta}_1 + l_2 \left(\ddot{\theta}_2 \cos(\theta_1 - \theta_2) + \dot{\theta}_2 (-\sin(\theta_1 - \theta_2)) (\dot{\theta}_1 - \dot{\theta}_2) \right) \right) \\ - \left(m_2 l_1 \dot{\theta}_1 l_2 \dot{\theta}_2 (-\sin(\theta_1 - \theta_2)) + g l_1 (-\sin \theta_1) (m_1 + m_2) \right) = 0 \end{aligned}$$

$$\begin{aligned} m_1 l_1^2 \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \left(\ddot{\theta}_2 \cos(\theta_1 - \theta_2) + \dot{\theta}_2 (-\sin(\theta_1 - \theta_2)) (\dot{\theta}_1 - \dot{\theta}_2) \right) \\ - m_2 l_1 \dot{\theta}_1 l_2 \dot{\theta}_2 (-\sin(\theta_1 - \theta_2)) - g l_1 (-\sin \theta_1) (m_1 + m_2) = 0 \end{aligned}$$

$$\begin{aligned} (m_1 + m_2) l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \\ + m_2 l_1 l_2 \dot{\theta}_2 (-\sin(\theta_1 - \theta_2)) (\dot{\theta}_1 - \dot{\theta}_2) \\ - m_2 l_1 l_2 \dot{\theta}_2 \dot{\theta}_1 (-\sin(\theta_1 - \theta_2)) \\ - g l_1 (-\sin \theta_1) (m_1 + m_2) = 0 \end{aligned}$$

$$\begin{aligned} (m_1 + m_2) l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) \\ + m_2 l_1 l_2 \dot{\theta}_2 (-\sin(\theta_1 - \theta_2)) \left((\dot{\theta}_1 - \dot{\theta}_2) - \dot{\theta}_1 \right) \\ - g l_1 (-\sin \theta_1) (m_1 + m_2) = 0 \end{aligned}$$

Euler–Lagrange equation for θ_1 :

$$\begin{aligned} (m_1 + m_2)l_1^2\ddot{\theta}_1 + m_2l_1l_2\ddot{\theta}_2\cos(\theta_1 - \theta_2) \\ = -gl_1(m_1 + m_2)\sin\theta_1 - m_2l_1l_2\dot{\theta}_2^2\sin(\theta_1 - \theta_2) \end{aligned}$$

For θ_2

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_2}\right) - \frac{\partial L}{\partial \theta_2} = 0$$

First $\partial L/\partial \theta_2$:

$$\begin{aligned} \frac{\partial L}{\partial \theta_2} &= \frac{\partial}{\partial \theta_2} \left[\frac{1}{2}m_1l_1^2\dot{\theta}_1^2 \right. \\ &\quad \left. + \frac{1}{2}m_2\left(l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + 2l_1l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)\right) \right. \\ &\quad \left. + g(m_1(y_0 + l_1\cos\theta_1) + m_2(y_0 + l_1\cos\theta_1 + l_2\cos\theta_2)) \right] \\ \frac{\partial L}{\partial \theta_2} &= \frac{1}{2}m_2\left(2l_1l_2\dot{\theta}_1\dot{\theta}_2(-\sin(\theta_1 - \theta_2) \cdot (-1))\right) + g(m_2l_2(-\sin\theta_2)) \\ \frac{\partial L}{\partial \theta_2} &= m_2l_1l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) - gm_2l_2\sin\theta_2 \end{aligned}$$

Then $\partial L/\partial \dot{\theta}_2$:

$$\begin{aligned} \frac{\partial L}{\partial \dot{\theta}_2} &= \frac{1}{2}m_2\left(2l_2^2\dot{\theta}_2 + 2l_1l_2\dot{\theta}_1\cos(\theta_1 - \theta_2)\right) \\ \frac{\partial L}{\partial \dot{\theta}_2} &= m_2l_2^2\dot{\theta}_2 + m_2l_1l_2\dot{\theta}_1\cos(\theta_1 - \theta_2) \\ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_2}\right) &= m_2l_2^2\ddot{\theta}_2 \\ &\quad + m_2l_1l_2\left(\ddot{\theta}_1\cos(\theta_1 - \theta_2) + \dot{\theta}_1(-\sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2))\right) \end{aligned}$$

Combining and simplifying:

$$\begin{aligned} m_2l_2^2\ddot{\theta}_2 + m_2l_1l_2\left(\ddot{\theta}_1\cos(\theta_1 - \theta_2) + \dot{\theta}_1(-\sin(\theta_1 - \theta_2)(\dot{\theta}_1 - \dot{\theta}_2))\right) \\ - m_2l_1l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) + gm_2l_2\sin\theta_2 = 0 \end{aligned}$$

$$\begin{aligned} m_2l_2^2\ddot{\theta}_2 + m_2l_1l_2\ddot{\theta}_1\cos(\theta_1 - \theta_2) \\ + m_2l_1l_2\dot{\theta}_1\left(-\dot{\theta}_1\sin(\theta_1 - \theta_2) + \dot{\theta}_2\sin(\theta_1 - \theta_2)\right) \\ - m_2l_1l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) + gm_2l_2\sin\theta_2 = 0 \end{aligned}$$

$$\begin{aligned} m_2l_2^2\ddot{\theta}_2 + m_2l_1l_2\ddot{\theta}_1\cos(\theta_1 - \theta_2) \\ - m_2l_1l_2\dot{\theta}_1^2\sin(\theta_1 - \theta_2) + m_2l_1l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) \\ - m_2l_1l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) + gm_2l_2\sin\theta_2 = 0 \end{aligned}$$

$$m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) - m_2 l_1 l_2 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + g m_2 l_2 \sin \theta_2 = 0$$

Euler–Lagrange equation for θ_2 :

$$m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) = m_2 l_1 l_2 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) - g m_2 l_2 \sin \theta_2$$

System of equations

$$\begin{cases} (m_1 + m_2) l_1^2 \ddot{\theta}_1 + m_2 l_1 l_2 \ddot{\theta}_2 \cos(\theta_1 - \theta_2) = -g l_1 (m_1 + m_2) \sin \theta_1 - m_2 l_1 l_2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) \\ m_2 l_1 l_2 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) + m_2 l_2^2 \ddot{\theta}_2 = m_2 l_1 l_2 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) - m_2 g l_2 \sin \theta_2 \end{cases}$$

$$\begin{cases} \ddot{\theta}_1 + \frac{m_2 l_1 l_2}{(m_1 + m_2) l_1^2} \ddot{\theta}_2 \cos(\theta_1 - \theta_2) = \frac{-g l_1 (m_1 + m_2)}{(m_1 + m_2) l_1^2} \sin \theta_1 - \frac{m_2 l_1 l_2 \dot{\theta}_2^2}{(m_1 + m_2) l_1^2} \sin(\theta_1 - \theta_2) \\ \ddot{\theta}_2 + \frac{m_2 l_1 l_2}{m_2 l_2^2} \ddot{\theta}_1 \cos(\theta_1 - \theta_2) = \frac{m_2 l_1 l_2 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2)}{m_2 l_2^2} - \frac{m_2 g l_2}{m_2 l_2^2} \sin \theta_2 \end{cases}$$

$$\begin{cases} \sigma_1 = \frac{m_2 l_1 l_2 \cos(\theta_1 - \theta_2)}{(m_1 + m_2) l_1^2} \\ \ddot{\theta}_1 + \sigma_1 \ddot{\theta}_2 = -\frac{g}{l_1} \sin \theta_1 - \frac{\sigma_1}{\cos(\theta_1 - \theta_2)} \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) \\ \sigma_2 = \frac{m_2 l_1 l_2 \cos(\theta_1 - \theta_2)}{m_2 l_2^2} \\ \ddot{\theta}_2 + \sigma_2 \ddot{\theta}_1 = \frac{l_1 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2)}{l_2} - \frac{\sigma_2}{\cos(\theta_1 - \theta_2)} \frac{g}{l_1} \sin \theta_2 \end{cases}$$

$$\begin{cases} \sigma_1 = \frac{m_2 l_1 l_2 \cos(\theta_1 - \theta_2)}{(m_1 + m_2) l_1^2} \\ \sigma_2 = \frac{m_2 l_1 l_2 \cos(\theta_1 - \theta_2)}{m_2 l_2^2} \\ f_1(x) = -\frac{g}{l_1} \sin \theta_1 - \frac{\sigma_1}{\cos(\theta_1 - \theta_2)} \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) \\ f_2(x) = \frac{l_1 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2)}{l_2} - \frac{\sigma_2}{\cos(\theta_1 - \theta_2)} \frac{g}{l_1} \sin \theta_2 \\ \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = \begin{pmatrix} 1 & \sigma_1 \\ \sigma_2 & 1 \end{pmatrix} \begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{pmatrix} \end{cases}$$

Now we need to check if that matrix is invertible

$$\begin{aligned} \det \begin{pmatrix} 1 & \sigma_1 \\ \sigma_2 & 1 \end{pmatrix} &= 1 - \sigma_1 \sigma_2 = 1 - \frac{m_2 l_1 l_2 \cos(\theta_1 - \theta_2)}{(m_1 + m_2) l_1^2} \frac{m_2 l_1 l_2 \cos(\theta_1 - \theta_2)}{m_2 l_2^2} \\ &= 1 - \frac{\cos(\theta_1 - \theta_2) m_2 \cos(\theta_1 - \theta_2)}{(m_1 + m_2) l_2^2} \\ &= 1 - \frac{m_2 \cos^2(\theta_1 - \theta_2)}{(m_1 + m_2)} \end{aligned}$$

First, we analyze the upper bounds of the individual components. Since physical masses are strictly positive ($m_1, m_2 > 0$), the mass ratio is strictly less than 1:

$$0 < \frac{m_2}{m_1 + m_2} < 1$$

Second, the square of the cosine function is naturally bounded by 1:

$$0 \leq \cos^2(\theta_1 - \theta_2) \leq 1$$

Multiplying these two terms guarantees that their product is strictly less than 1:

$$\frac{m_2 \cos^2(\theta_1 - \theta_2)}{m_1 + m_2} < 1$$

Therefore, subtracting this product from 1 yields a strictly positive result:

$$1 - \frac{m_2 \cos^2(\theta_1 - \theta_2)}{m_1 + m_2} > 0$$

This means that **the matrix is invertible**. We can do this directly:

$$A = \begin{pmatrix} 1 & \sigma_1 \\ \sigma_2 & 1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A} \begin{pmatrix} 1 & -\sigma_1 \\ -\sigma_2 & 1 \end{pmatrix} = \frac{1}{1 - \sigma_1 \sigma_2} \begin{pmatrix} 1 & -\sigma_1 \\ -\sigma_2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = \begin{pmatrix} 1 & \sigma_1 \\ \sigma_2 & 1 \end{pmatrix} \begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{pmatrix}$$

$$\begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{pmatrix} = \begin{pmatrix} 1 & \sigma_1 \\ \sigma_2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$$

$$\begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{pmatrix} = \frac{1}{1 - \sigma_1 \sigma_2} \begin{pmatrix} 1 & -\sigma_1 \\ -\sigma_2 & 1 \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$$

$$\begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{pmatrix} = \frac{1}{1 - \sigma_1 \sigma_2} \begin{pmatrix} f_1 - \sigma_1 f_2 \\ f_2 - \sigma_2 f_1 \end{pmatrix}$$

So now we can write a system of differential equations for a double pendulum.

$$\omega_1 = \dot{\theta}_1 \quad \omega_2 = \dot{\theta}_2$$

$$\left\{ \begin{array}{l} \sigma_1 = \frac{m_2 l_1 l_2 \cos(\theta_1 - \theta_2)}{(m_1 + m_2) l_1^2} \\ \sigma_2 = \frac{m_2 l_1 l_2 \cos(\theta_1 - \theta_2)}{m_2 l_2^2} \\ f_1(x) = -\frac{g}{l_1} \sin \theta_1 - \frac{\sigma_1}{\cos(\theta_1 - \theta_2)} \omega_2^2 \sin(\theta_1 - \theta_2) \\ f_2(x) = \frac{l_1}{l_2} \omega_1^2 \sin(\theta_1 - \theta_2) - \frac{\sigma_2}{\cos(\theta_1 - \theta_2)} \frac{g}{l_1} \sin \theta_2 \\ \frac{d}{dt} \begin{pmatrix} \theta_1 \\ \theta_2 \\ \omega_1 \\ \omega_2 \end{pmatrix} = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \frac{f_1 - \sigma_1 f_2}{1 - \sigma_1 \sigma_2} \\ \frac{f_2 - \sigma_2 f_1}{1 - \sigma_1 \sigma_2} \end{pmatrix} \end{array} \right.$$